NES-VMC 算法的核心思想正是将原系统前K个激发态的求解问题，等价地转化为一个“扩展系统”的基态求解问题。 文献链接:https://arxiv.org/abs/2308.16848 以下是对 H2 分子系统为例子的实现 基于 Netket 与 Flax.nnx 框架 首先`hi_K = nk.hilbert.TensorHilbert(hi**K)`是指的拓展 Hilbert 空间。 原系统的 Hilbert 空间是 `hi`，拓展后的 Hilbert 空间是 `hi_K`。 `hi.all_states()` 每个组态是 (4,) 的向量，对应着 [α0 α1 β0 β1]-> x_1。 `hi_K.all_states()` 每个组态是 (4*K,) 的向量，对应着 $x^1,x^2,...x^k$。 例如 K=2 的时候 `hi_K.all_states()[0]` `>>Array([0, 1, 0, 1, 0, 1, 0, 1], dtype=int8)` 对应着 $x^1=[0,1,0,1],x^2=[0,1,0,1]$ 原系统共有 4 个合法组态 `[0, 1, 0, 1]`,`[0, 1, 1, 0]`,`[1, 0, 0, 1]`,`[1, 0, 1, 0]`, 组态顺序是：[α0 α1 β0 β1] 然而为了能够顺利地进行 NES-VMC 采样，我们需要一次性采样K个样本。以 K=2 为例子, 每次我们需要采样2个组态. 在我的例子中 我使用了`sampler = nk.sampler.MetropolisSampler(hi_ensemble, rule=tensor_rule, n_chains=16, sweep_size=32)`

```
import jax
import jax.numpy as jnp
import netket as nk
import netket.experimental as nkx
import numpy as np
from pyscf import gto, scf, fci
from flax import nnx

#==============================================================================
# 1. 全局参数 & H₂ 分子定义
#==============================================================================
K = 2  # NES-VMC 要计算的低激发态数量（基态+1个激发态）
bond_length = 1.4
geometry = [('H', (0., 0., 0.)), ('H', (bond_length, 0., 0.))]
mol = gto.M(atom=geometry, basis='STO-3G', verbose=0)
mf = scf.RHF(mol).run(verbose=0)

# FCI 精确基准（4个态）
cisolver = fci.FCI(mf)
cisolver.nroots = 4
E_fcis, fcivec = cisolver.kernel()
print("="*60)
print("H₂ FCI 基准能量")
print("="*60)
for i, e in enumerate(E_fcis):
    exc = (e - E_fcis[0]) * 27.2114
    print(f"E{i} = {e:.8f} Ha  |  激发能: {exc:.4f} eV")

# 转为 NetKet 哈密顿量
ha = nkx.operator.from_pyscf_molecule(mol)

# 1. 定义原始 Hilbert 空间（同之前）
hi = nk.hilbert.SpinOrbitalFermions(
    n_orbitals=2,
    s=1/2,
    n_fermions_per_spin=(1,1),
)
# 2. 定义 NES-VMC 参数
K = 2  # 想要同时采样的组态数
# 3. 修正点：使用指数运算符 ** 来构建扩展 Hilbert 空间
hi_ensemble = hi ** K
# 4. 后续的采样器设置保持不变
edges = [(0, 1), (2, 3)] #edges 是对 hi 的 edges
g = nk.graph.Graph(edges=edges)
# hi_ensemble.all_states()[0].shape # (12,)

hi_K = nk.hilbert.TensorHilbert(hi**K)
hi_K.all_states().shape #(4**K,4*K)

single_rule = nk.sampler.rules.FermionHopRule(hilbert=hi, graph=g)
rules_tuple = tuple(single_rule for _ in range(K))
tensor_rule = nk.sampler.rules.TensorRule(hilbert=hi_K, rules=[single_rule]*K)
sampler = nk.sampler.MetropolisSampler(hi_ensemble, rule=tensor_rule, n_chains=16, sweep_size=32)
```

接下来要设计原文中对应的Single-State Ansatz $\psi(x)$ 需要注意的是 Netket 里的采样器接受的只能是 $log(\psi(x))$，不能是 $\psi(x)$。 NESTotalAnsatz 对应着原文中的 $\Psi(x^1,x^2,...,x^k) = \det(M)$ $$\Psi(\mathbf{x}^1, \dots, \mathbf{x}^K) \triangleq \det \begin{pmatrix} \psi_1(\mathbf{x}^1) & \dots & \psi_K(\mathbf{x}^1) \ \vdots & & \vdots \ \psi_1(\mathbf{x}^K) & \dots & \psi_K(\mathbf{x}^K) \end{pmatrix} $$

```
class SingleStateAnsatz(nnx.Module):
    def __init__(self, n_spin_orbitals: int, hidden_dim=16, *, rngs: nnx.Rngs):
        super().__init__()
        self.linear1 = nnx.Linear(n_spin_orbitals, hidden_dim, rngs=rngs, param_dtype=complex)
        self.linear2 = nnx.Linear(hidden_dim, hidden_dim, rngs=rngs, param_dtype=complex)
        self.output = nnx.Linear(hidden_dim, 1, rngs=rngs, param_dtype=complex)

    def __call__(self, x):
        h = nnx.tanh(self.linear1(x.astype(complex)))
        h = nnx.tanh(self.linear2(h))
        out = self.output(h)
        #log_out = jnp.log(out)
        return jnp.squeeze(out)

class NESTotalAnsatz(nnx.Module):
    def __init__(self, n_spin_orbitals, n_states=K, hidden_dim=16):
        super().__init__()
        self.n_states = n_states
        self.n_spin_orbitals = n_spin_orbitals

        # 1. 生成一串不同的 JAX 随机 key
        key = jax.random.key(42)          # 主种子
        keys = jax.random.split(key, n_states)  # 拆成 n_states 个不同 key

        # 2. 给每个 ansatz 分配不同的 key → 不同的 nnx.Rngs
        # ✅ 这是 NNX 官方支持的写法
        self.single_ansatz_list = [
            SingleStateAnsatz(
                n_spin_orbitals,
                hidden_dim,
                rngs=nnx.Rngs(keys[i])  # 每个用不同 key
            )
            for i in range(n_states)
        ]
    def __call__(self, x_batch):
        '''
        x_batch: (K, n_spin_orbitals)
        '''
        K_state = self.n_states
        M = []
        for i in range(K_state):
            row = [self.single_ansatz_list[j](x_batch[i]) for j in range(K_state)]
            M.append(jnp.array(row))
        M = jnp.stack(M)
        psi_total = jnp.linalg.det(M)
        return psi_total, M
```

由于原文里的局域能量计算$E_L(\mathbf{x}) \triangleq \Psi^{-1}(\mathbf{x}) \hat{H} \Psi(\mathbf{x})$, 其中$\hat{H}$是(4,4)维的哈密顿量矩阵. $$\hat{H}\Psi(\mathbf{x}) \triangleq \begin{pmatrix} \hat{O}\psi_1(\mathbf{x}^1) & \dots & \hat{O}\psi_K(\mathbf{x}^1) \ \vdots & & \vdots \ \hat{O}\psi_1(\mathbf{x}^K) & \dots & \hat{O}\psi_K(\mathbf{x}^K) \end{pmatrix}$$

为了计算$\hat{H}\Psi(\mathbf{x})$，我们需要先计算$\psi(\mathbf{x})$，再用$\hat{H}$的每个元素乘以$\psi(\mathbf{x})$，最后求和。 基于以下两个函数运算。get_conn_padded 是 Netket 中支持 jax.jit的版本

```
def Ham_psi(ha: nk.operator.DiscreteOperator, single_ansatz:SingleStateAnsatz, x: jnp.array):
    x_primes, mels = ha.get_conn_padded(x)
    psi_vals = jax.vmap(single_ansatz)(x_primes)
    return jnp.sum(mels * psi_vals)

def Ham_Psi(ha:nk.operator.DiscreteOperator, total_ansatz: NESTotalAnsatz, x_batch):
```

```
    K_state = total_ansatz.n_states
    H_mat = []
    for i in range(K_state):
        row = []
        for j in range(K_state):
            v = Ham_psi(ha, total_ansatz.single_ansatz_list[j], x_batch[i])
            row.append(v)
        H_mat.append(row)
    return jnp.array(H_mat, dtype=complex)
```

计算局域能量的相关函数

samples.shape = (n_chain, chain_length, n_spin_orbitals*K) 所以往往要把 samples.reshape(-1,K,n_spin_orbitals) 来给这些函数使用

```
def compute_local_energy(ha:nk.operator.DiscreteOperator, total_ansatz: NESTotalAnsatz, x_batch):
    """
    x_batch.shape = (K, n_spin_orbitals)
    """
    psi, M = total_ansatz(x_batch)
    eps = 1e-5
    M = M + eps * jnp.eye(M.shape[0], dtype=M.dtype)
    Hp = Ham_Psi(ha, total_ansatz, x_batch)
    el_mat = jnp.linalg.solve(M, Hp)
    return jnp.trace(el_mat), el_mat

def compute_local_energy_single(ham: nk.operator.DiscreteOperator,
                                model: NESTotalAnsatz,
                                x_batch):
    # x_batch: (K, n_spin_orbitals)
    tr_el, el_mat = compute_local_energy(ham, model, x_batch)
    return tr_el.real, el_mat  # 确保返回实数迹


compute_local_energy_batch = jax.vmap(
    compute_local_energy_single,
    # 对应 3 个入参: ham, model, x_batch
    # x_batch.shape = (batch_size,K, n_spin_orbitals)
    in_axes=(None, None, 0),
    out_axes=(0, 0),
    axis_name='samples_batch'
)

def compute_average_local_energy(ham: nk.operator.DiscreteOperator,
                                 model: NESTotalAnsatz,
                                 samples,
                                 ):

    '''
    samples.shape = (n_samples, K, n_spin_orbitals)
    '''
    tr_els, el_mats = compute_local_energy_batch(
        ham, model, samples
    )
    tr_avg = tr_els.mean()
    el_mat_avg = el_mats.mean(axis=0)
    return tr_avg, el_mat_avg

def loss_fn_single(parameters, ham, x_batch):
    # 把参数放回模型
    graph,params = parameters
    model = nnx.merge(graph, params)
    # 计算 loss = trace(local energy matrix)
    tr_el, _ = compute_local_energy_single(ham, model, x_batch)
    return tr_el  # 我们对这个标量求导!

def loss_fn_single(parameter, ham, x_batch):
    # 把参数放回模型
    graph,params = parameters
    model = nnx.merge(graph, params)
    # 计算 loss = trace(local energy matrix)
```

```
    tr_el, _ = compute_local_energy_single(ham, model, x_batch)
    return tr_el  # 我们对这个标量求导!


def loss_fn_batch(params,ham, x_batch):
    # 把参数放回模型
    graphdef, variables = params
    model = nnx.merge(graphdef, variables)
    # 计算 loss = trace(local energy matrix)
    tr_el, _ = compute_average_local_energy(ham, model, x_batch)
    return tr_el  # 我们对这个标量求导!

value_and_grad_single = jax.value_and_grad(loss_fn_single)
value_and_grad_batch = jax.value_and_grad(loss_fn_batch)
```

samples.shape = (n_chain, chain_length, n_spin_orbitals*K) 所以往往要把 samples.reshape(-1,K,n_spin_orbitals) 来给这些函数使用 forward 函数要满足 f(parameters, σ) -> jax.Array.

```
def forward(params, x_batch):
    #print(x_batch.shape)
    # x_batch: (n_chains, chain_length, n_spin_orbitals*K)
    n_chains = x_batch.shape[0]
    K = 2
    n_spin = 4
    # 重塑为 (n_chains, K, n_spin)
    x_reshaped = x_batch.reshape(n_chains, K, n_spin)

    # 定义单个联合样本的计算
    def single_logpsi(params, x):
        (psi, _), _ = nnx.call(params)(x)

        return jnp.log(jnp.abs(psi))  # 标量复数

    # 批量映射
    log_psi_batch = jax.vmap(single_logpsi, in_axes=(None, 0))(params, x_reshaped)
    return log_psi_batch  # 形状 (n_chains,)

parameters = nnx.split(total_ansatz)
sampler_state = sampler.init_state(forward, parameters, seed=1)
samples, sampler_state = sampler.sample(
    forward, parameters, state=sampler_state, chain_length=500
)
samples.shape = (n_chain, chain_length, n_spin_orbitals*K)
```

最后的训练过程 我认为要在每次循环过程中 先reset sampler_state,再生成样本 reset sampler-> 生成新样本 -> 基于新样本 计算loss、梯度 -> 更新model的参数 ->reset sampler...

```
total_ansatz = NESTotalAnsatz(4,2,4*4)
graphdef, variables = nnx.split(total_ansatz)  # ✅ 拆成结构 + 参数
sampler_state = sampler.init_state(forward, (graphdef, variables), seed=1)
# 优化器 (Adam 稳定)
optimizer = optax.adam(learning_rate=1e-2)
opt_state = optimizer.init(variables)  # ✅ 只初始化变量! 不传 graphdef!
loss_record=[]
for step in range(100):
    sampler_state = sampler.reset(forward, (graphdef, variables), sampler_state)
    samples, sampler_state = sampler.sample(
        forward, (graphdef, variables), state=sampler_state, chain_length=200
    )
    params_for_loss = (graphdef, variables)
    loss, grads = value_and_grad_batch(
        params_for_loss, ha, samples.reshape(-1, K, 4)
    )
    loss_record.append(loss)
    if step % 20 == 0:
        print(f'Trace of Total_ansatz matirx| [{step}] loss: {loss:.8f} ')
    grad_graph, grad_vars = grads
    updates, opt_state = optimizer.update(grad_vars, opt_state, variables)
```

```
    variables = optax.apply_updates(variables, updates)
    total_ansatz = nnx.merge(graphdef, variables)
```

获得计算得到的基态、激发态能量

```
value, M = compute_average_local_energy(ha, total_ansatz, samples_final.reshape(-1,2,4))
eigen_energies = jnp.linalg.eigvalsh(M).real
eigen_energies
```

这是一个 Netket 官方给出的一个训练案例:

```
# Settings
model = Jastrow()  # Try both MF() and Jastrow()
sampler = nk.sampler.MetropolisSampler(
    hi,
    nk.sampler.rules.LocalRule(),
    n_chains=20
)
n_iters = 300
chain_length = 1000 // sampler.n_chains

# Initialize
parameters = model.init(jax.random.key(0), np.ones((hi.size,)))
sampler_state = sampler.init_state(model, parameters, seed=1)

# Logging
logger = nk.logging.RuntimeLog()

for i in tqdm(range(n_iters)):
    # sample
    sampler_state = sampler.reset(model, parameters, state=sampler_state)
    samples, sampler_state = sampler.sample(model, parameters, state=sampler_state, chain_length=chain_length

    # compute energy and gradient
    E, E_grad = estimate_energy_and_gradient(model, parameters, hamiltonian_jax, samples)

    # update parameters
    parameters = jax.tree.map(lambda x,y: x-0.005*y, parameters, E_grad)

    # log energy
    logger(step=i, item={'Energy':E})
```